



Why AI Security Starts with Data and Why NYC Agencies Should Care

If your AI is trained on bad data, it makes bad decisions. That’s not just inconvenient—it can be dangerous, especially for public sector agencies. This guide breaks down critical security strategies from the NSA and CISA that help ensure your data (and your AI) stay safe, trustworthy, and effective.

THE BIG RISKS

Data Poisoning
Hackers insert false info into your datasets

Data Drift
Over time, models degrade as real-world data changes

Bias & Errors
Unintended or malicious patterns in data cause bad outcomes

WHO IS THIS FOR

City and state [agency leaders](#) using or evaluating AI tools

CIOs, CISOs, [compliance officers](#) and procurement teams

Public [tech contractors](#) needing to meet federal AI data standards

AI Lifecycle & Threats Explained Simply

Where AI Data Security Can Fail – And How to Prevent It

Lifecycle Phase	What To Focus On	Threats to Watch For
Plan & Design	Threat modeling, privacy-by-design	Bad sourcing, insecure planning
Collect & Process	Validate, sanitize, and encrypt data	Poisoned data, insider attacks
Build & Train Model	Secure training pipelines & logs	Poisoned data, model theft
Verify & Validate	Run adversarial tests, integrity checks	Poisoned inputs, biased output
Deploy & Use	Encrypt APIs, monitor activity	Data drift, unauthorized access
Operate & Monitor	Log access, audit regularly	Breach, bias, drift, bad updates

 Any failure in these stages can lead to:

- Compromised decision-making
- Privacy violations (PII leaks)
- Legal/regulatory exposure
- Erosion of public trust

10-Step NSA/CISA Data Security Checklist

Each step is rewritten in plain English while preserving its original meaning

1 Use Trusted Data Only

No scraping. Only source from verified, reputable providers

2 Track Every File's Origin

Use tools to log who, when, and where data was collected or changed

3 Verify It Hasn't Been Tampered With

Use cryptographic hashes to detect any file changes

4 Digitally Sign Your Data

So you know who made edits and when—use signatures you can trust

5 Lock Down Storage & Transfers

Encrypt everything in transit and at rest (AES-256 or better)

6 Go Zero Trust

No open doors: use secure enclaves and strict user controls

7 Classify Everything

Know what data is sensitive and limit who sees what

8 Delete Like You Mean It

Use certified, permanent deletion tools—no simple “delete” buttons

9 Add Privacy Shields

Use depersonalization and techniques like differential privacy

10 Audit and Adjust Continuously

Run regular assessments. Update your models and controls

This guidance is based entirely on the NSA/CISA Cybersecurity Information Sheet: “AI Data Security” (May 2025), co-authored with international partners. SJ Innovation has reformatted it for clarity and awareness building only.